GenMol / UDLM study through the paired objective evaluation

Completed V5/V6 screens, V7 resource pilot, V8 incident audit, separate V8b CE training and V9 molecular results

No superiority established. The highest observed repaired quality among the 22 disclosed engineering configurations is 57.031% (V6, s_t050_gibbs); contextual local MDLM quality is 85.8% and published GenMol V1 quality is 84.6%. This maximum is selected from small exploratory screens. A higher pilot mean, if present, does not establish superiority or authorize a final candidate promotion.

V9 clean CE minus CT repaired quality. primary at T=1.0: -5.500 percentage points; secondary at T=0.5: -2.000 percentage points. These are descriptive paired-seed differences; all strict/repaired metrics and every configuration remain disclosed below.

Study	Question and complete scope	Interpretation
V5	R/S/E x four temperatures; 24 runs / 1,536 requests	Historical 1,000-update batch-16 CT models; temperature exploration
V6	R/S/E x predictor/Gibbs; 12 runs / 768 requests	128 predictors versus 64 predictors + 64 Gibbs evaluations; mixed gains
V7	20-update batch-128 CT-E throughput pilot; 2,560 exposures	Resource feasibility only; no molecular evaluation or warm-start reuse
V8 / V8b	Separate 1,000-update CT and CE adaptations; 128,000 configured exposures each	Original CT controller/campaign failed; CT accepted separately; new CE controller completed
V9	CT/CE x temperatures 1.0 / 0.5; 8 runs / 800 requests	Primary CE minus CT at 1.0; informed secondary contrast at 0.5
Local MDLM	50,000 updates; 3 seeds / 3,000 requests	Contextual baseline; training/inference budgets and sample counts differ

New evidence and historical context. All 3,104 V5/V6/V9 molecular requests were independently rescored within their original runs; they are not pooled into a single model estimate. The historical 27-page report is appended without modifying its original bytes. Its CE-untrained sentence describes the earlier snapshot and is superseded by the completed V8b/V9 evidence here. Archived failure states remain failures.

Metrics. Validity = valid/requested; uniqueness = unique/valid; quality = unique valid molecules with QED >= 0.6 and SA <= 4, divided by requests; diversity is the released PyTDC fingerprint metric on unique valid molecules. Repaired decoding uses SAFE fix=True and largest-component selection by SMILES length; strict decoding uses fix=False and RDKit validation. Means and sample SD weight seeds equally; two-seed SD is not a confidence interval. Diversity and configurations are not pooled.

Training, controller outcomes and checkpoint semantics

Run	Updates / global batch / seed / GPUs	Training subprocess seconds	Accepted exposures / examples per second	Observed memory MiB
V7	20 / 128 / 1400 / 1	146.333	2,560 / 17.494	9637
V8_CT	1000 / 128 / 1500 / 2	785.822	not accepted by failed controller	9439, 11035
V8b_CE	1000 / 128 / 1500 / 2	706.595	128,000 / 181.150	11139, 9273

V8 CT incident. CT reached step 1,000 and exited zero, but the controller immediately found a residual distributed process group and failed, retaining both leases. CE never started in that campaign. A separately hashed CPU audit later verified the saved CT checkpoint and absent processes, then released the exact retained leases. It supports 128,000 configured exposures; the original accepted-exposure and throughput fields remain null. The exact transient group members were not captured, so the teardown-race explanation remains an inference.

Separate V8b CE run. Fresh MDLM EMA initialization preserved the original CE settings. A repaired controller allowed up to 15 seconds for teardown; its group exited after 0.751 seconds with 4 probes. Its own completion receipt accepted step 1,000, optimizer state, EMA updates, finite saved tensors and the CE marker/metadata. This does not convert the failed V8 campaign into a success.

Matched training. CT and CE each start fresh from MDLM 50k EMA, seed 1500, global batch 128 = 2 GPUs x microbatch 16 x accumulation 4; 1,000 updates = 128,000 configured exposures per arm. Both share the empirical prior with 0.0002 uniform mixture, FiLM, L1 schedule, full corruption alphabet and all-special-token clean-target mask. V7 and the earlier V5/V6 models used historical masks; V9 is not an isolated continuation of those screens.

Timing limits. Training-subprocess time includes startup and checkpoint saving. Short V7 startup overhead, one versus two GPUs, sequential runs and external processes prevent a controlled speedup claim. Memory values are two-second external aggregate GPU observations including other processes, not allocator peaks. Exposures are configured accounting, not distinct molecules or an independent live row trace. Saved-tensor finiteness does not prove every intermediate update was finite. CT/CE loss magnitudes are not comparable quality scores.

Checkpoint	SHA-256
V8 CT: separately audited	48986899c401c09cdc1e9e865e773cadc40899b62129420689f89a8e622a9c99
V8b CE: completed separately	b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1
Common warm-start MDLM 50k EMA	8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6

V9: all four objective settings

All V9 rows use 200 requests (two 100-request seeds, 1600/1601), EMA weights, 128 predictor evaluations, no Gibbs corrector, top-p 1.0, endpoint 1e-5, randomness 0 and min_add_len 40. CE clean logits are converted to LOO at the current noisy state/time before temperature and top-p; CT already predicts raw LOO. The conversion adds no backbone evaluation. Temperature 1.0 is primary; 0.5 was informed by V5/V6.

Config	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t100	repaired	97.500 +/- 2.121	99.479 +/- 0.737	46.500 +/- 6.364	0.909 +/- 0.000
ct_t100	strict	39.000 +/- 5.657	100.000 +/- 0.000	21.000 +/- 1.414	0.886 +/- 0.007
ce_t100	repaired	98.000 +/- 1.414	99.485 +/- 0.729	41.000 +/- 4.243	0.903 +/- 0.004
ce_t100	strict	64.500 +/- 7.778	100.000 +/- 0.000	26.000 +/- 7.071	0.887 +/- 0.005
ct_t050	repaired	98.500 +/- 0.707	99.495 +/- 0.714	54.500 +/- 2.121	0.884 +/- 0.003
ct_t050	strict	77.000 +/- 1.414	100.000 +/- 0.000	44.000 +/- 1.414	0.868 +/- 0.008
ce_t050	repaired	99.000 +/- 0.000	99.495 +/- 0.714	52.500 +/- 4.950	0.888 +/- 0.006
ce_t050	strict	67.000 +/- 5.657	100.000 +/- 0.000	39.500 +/- 3.536	0.870 +/- 0.001

V9: paired CE minus CT effects

Positive deltas favor CE for that metric. Validity/uniqueness/quality deltas are percentage points; diversity uses its original scale. Values are the equal-seed mean +/- sample SD. An undefined metric withholds its summary until both declared seeds define it. The appendix retains each per-seed difference.

Contrast / T	Branch	Delta validity	Delta uniqueness	Delta quality	Delta diversity
primary / 1.0	repaired	0.500 +/- 0.707	0.005 +/- 0.008	-5.500 +/- 2.121	-0.006 +/- 0.004
primary / 1.0	strict	25.500 +/- 2.121	0.000 +/- 0.000	5.000 +/- 5.657	0.000 +/- 0.012
secondary / 0.5	repaired	0.500 +/- 0.707	0.000 +/- 1.428	-2.000 +/- 2.828	0.004 +/- 0.003
secondary / 0.5	strict	-10.000 +/- 7.071	0.000 +/- 0.000	-4.500 +/- 4.950	0.002 +/- 0.009

Seed pairing is not molecule-level matching: objectives and sampling paths differ. Both contrasts remain disclosed even when negative. The fixed design predates these checkpoint molecular outputs; earlier engineering results were already known, so this is not independent confirmation. Higher diversity alone does not imply higher validity or molecular quality. The appendix reports every requested metric under both decoding paths.

Reading the appendices. The first appendix is the complete paired V9 report with original per-run configuration, raw/rescore identities, CE helper provenance, generation counts and contrasts. The second is the preserved V5/V6 plus baseline overview, including all 18 historical settings and Gibbs ablations. The companion all_configurations.csv contains all 22 configuration means and SDs. The input manifest hashes the original reports, raw/run artifacts, training receipts/logs, CT audit and this generator. Large checkpoint identities are inherited from verified training/generation records; this overview does not reload the weights.

Equal updates/examples and shared hyperparameters do not imply equal compute or separately optimized objectives. All adaptations inherit the large MDLM pretraining budget; a matched extra-update MDLM control remains absent. Final UDLM seeds 0/1/2 remain reserved. No automatic promotion, further training or superiority declaration follows this report.